# Chapter 27

# Cheatsheets

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One-page topical summaries, aggregated here as a printable reference. Each is a self-contained reminder of the syntax, the assumptions, and the diagnostics that go with the methods it covers.

## 27.1 Foundations

### 27.1.1 Toolchain, tidy data, and `ggplot2`

#### 27.1.1.1 The research workflow

`Question→Measurements→Design→Acquisition→Description→Analysis→Interpretation→Validation→Knowledge`

Biological vs statistical significance: "$p < 0.05$" $\neq$ "effect matters".

## 27.2 R / RStudio / Quarto / renv

| Task | Command |
|---|---|
| Install packages | `install.packages("pkg")` |
| Pin project | `renv::init()` / `renv::snapshot()` / `renv::restore()` |
| Render one file | `quartorenderpath/to/file.qmd` |
| Preview with live reload | `quartopreview` |
| Render only slides | `quartorenderfile.qmd--torevealjs` |

## 27.3 Data types and scales

| Scale | Example | Statistic |
|---|---|---|
| Nominal | sex, site | counts, proportions |
| Ordinal | pain 0–10 | median, IQR, rank tests |
| Interval | °C | mean, SD |
| Ratio | kg, time | mean, SD, ratios |

Accuracy ≠ precision. Bias shifts the target; variance spreads the shots.

## 27.4 Tidy data

1. One variable per column.
2. One observation per row.
3. One observational unit per table.

## 27.5 `dplyr` verbs

```
df |>
  filter(age >= 18) |>
  select(id, age, sbp) |>
  mutate(sbp_z = (sbp - mean(sbp)) / sd(sbp)) |>
  group_by(arm) |>
  summarise(mean_sbp = mean(sbp, na.rm = TRUE), n = n())
```

Joins: `left_join`, `inner_join`, `anti_join` by a key.

Missingness: `is.na()`, `drop_na()`, `tidyr::complete()`.

## 27.6 `ggplot2` grammar

```
ggplot(data, aes(x, y, colour = group)) +
  geom_point() +
  facet_wrap(~ factor) +
  labs(x = "x label", y = "y label") +
  theme_minimal()
```

Layer order: `data→aes→geom→facet→scales→labels→theme`.

## 27.7 Plot families

- One variable continuous → **histogram**, **density**.

- Two continuous → **scatter**, sometimes binned / smoothed.
- One categorical + one continuous → **boxplot**, **violin**, **strip**.
- Two categorical → **bar with fill**, **mosaic**.

## 27.8 Decision rule

- Cannot read the data into a tibble → fix data entry first.
- Tibble reads but has missing or mixed types → clean before modelling.
- Plot is ugly or cluttered → trust the plot, not the summary statistic.

## 27.9 Common pitfalls

- Running statistics on untyped / partially-missing data.
- Adjusting colour scales instead of simplifying the chart.
- Reporting mean + SD for a skewed variable.
- Forgetting `set.seed()` in any simulation chunk.

## 27.10 Further reading

- Wickham, *R for Data Science* (2e).
- Posit ggplot2 and dplyr cheatsheets.

### 27.10.1 Descriptive statistics, Bayes, and distributions

#### 27.10.1.1 Descriptive statistics

| Situation | Summary |
|---|---|
| Roughly symmetric, no outliers | mean ± SD |
| Skewed or has outliers | median, IQR, range |
| Categorical | counts and proportions |
| Grouped | Table 1 via `gtsummary::tbl_summary()` |

## 27.11 Contingency tables

```r
table(df$x, df$y)
prop.table(table(df$x, df$y), margin = 1)
```

## 27.12 Bayes' theorem

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

In odds form: **posterior odds = prior odds × likelihood ratio.**

## 27.13 Diagnostic testing

| Metric | Formula |
|---|---|
| Sensitivity | TP / (TP + FN) |
| Specificity | TN / (TN + FP) |
| PPV | TP / (TP + FP) |
| NPV | TN / (TN + FN) |
| LR+ | Se / (1 − Sp) |
| LR− | (1 − Se) / Sp |

PPV collapses at low prevalence — even a 95%-accurate test is mostly false positives.

## 27.14 Discrete distributions

| Distribution | Use | R |
|---|---|---|
| Bernoulli($p$) | single trial | `rbinom(n,1,p)` |
| Binomial($n$, $p$) | # successes | `dbinom/pbinom/rbinom` |
| Poisson($\lambda$) | rare events | `dpois/ppois/rpois` |
| Negative binomial | overdispersed counts | `MASS::rnegbin`, `dnbinom` |

Binomial → Poisson as $n$ grows and $p$ shrinks with $np = \lambda$.

## 27.15 Continuous distributions

| Distribution | Use |
|---|---|
| Normal($\mu$, $\sigma$) | almost everything via the CLT |
| Student $t(\nu)$ | inference about means, small $n$ |
| $\chi^2(\nu)$ | variance, counts, GoF |
| F($\nu_1$, $\nu_2$) | variance ratios, ANOVA |
| Exponential($\lambda$) | time between events, memoryless |

R uses the `d/p/q/r` prefix: density, CDF, quantile, random.

## 27.16 Q-Q plot

```
ggplot(df, aes(sample = x)) + stat_qq() + stat_qq_line()
```

S-shape → heavy tails. U-shape → skew. Plot before the test.

## 27.17 Decision rule

- Reporting a mean? First check a histogram.
- Testing a proportion? First sketch a 2×2 table.
- Choosing a distribution? Simulate before believing.

## 27.18 Common pitfalls

- Quoting PPV without disclosing prevalence.
- Assuming normality because $n > 30$.
- Using mean ± SD on skewed data.

## 27.19 Further reading

- Altman, *Practical Statistics for Medical Research*.
- Gelman et al., *Bayesian Data Analysis*, ch. 1.

### 27.19.1 Sampling, estimation, and one-sample tests

#### 27.19.1.1 Populations vs samples

- **Parameter** = truth about the population ($\mu$, $\sigma$, $\pi$).
- **Estimator** = statistic computed on a sample ($\bar{x}$, $s$, $\hat{p}$).
- **Bias** = $E[\hat{\theta}] - \theta$. **MSE** = bias$^2$ + variance.

## 27.20 Central limit theorem

For iid samples with finite variance, as $n \to \infty$: $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$

Holds roughly by $n \approx 30$ for most non-pathological distributions.

## 27.21 Standard error

$\text{SE}(\bar{X}) = \sigma/\sqrt{n}$, estimated by $s/\sqrt{n}$.

## 27.22 Bootstrap

```r
B <- 2000
boot_mean <- replicate(B, mean(sample(x, replace = TRUE)))
quantile(boot_mean, c(0.025, 0.975))   # 95% percentile CI
sd(boot_mean)                           # bootstrap SE
```

## 27.23 Permutation test

```r
obs  <- mean(a) - mean(b)
pool <- c(a, b)
null <- replicate(5000, {
  idx <- sample(seq_along(pool), length(a))
  mean(pool[idx]) - mean(pool[-idx])
})
mean(abs(null) >= abs(obs))   # two-sided p
```

## 27.24 Maximum likelihood

Given data $x$ and model $f(x;\theta)$:

- $\ell(\theta) = \sum \log f(x_i;\theta)$.
- $\hat{\theta}_{\text{MLE}} = \arg\max_\theta \ell(\theta)$.
- Fisher information $I(\theta)$; asymptotic SE $= 1/\sqrt{I(\hat{\theta})}$.

## 27.25 One-sample tests (five-step template)

**Hypothesis → Visualise → Assumptions → Conduct → Conclude.**

| Test | R | Assumptions |
|---|---|---|
| One-sample $t$ | `t.test(x,mu=0)` | roughly normal or large $n$ |
| One-proportion | `prop.test(k,n,p=0.5)` or `binom.test` | $np, n(1-p) \geq 5$ for `prop.test` |

## 27.26 Hypothesis-testing vocabulary

- **Type I**: reject true $H_0$ (probability $\alpha$).
- **Type II**: accept false $H_0$ (probability $\beta$).
- **Power** $= 1 - \beta$.
- A $p$-value is **not** the probability that $H_0$ is true.

## 27.27 Decision rule

- Want a CI? Bootstrap first, then ask whether a formula applies.
- Want a test? State $H_0$ and $\alpha$ in writing before running it.
- Want to know if $n$ is big enough? Simulate the CLT for your outcome.

## 27.28 Common pitfalls

- Reporting SE when you meant SD (or vice versa).
- Using the 95% CI to "prove" there is no effect.
- Running many tests and quoting the smallest $p$-value.

## 27.29 Further reading

- Harrell, *Biostatistics for Biomedical Research*, ch. 3–5.
- Efron & Tibshirani, *An Introduction to the Bootstrap*.

### 27.29.1 Two-group comparisons, effect sizes, and power

#### 27.29.1.1 Choosing a comparison

| Design | Continuous outcome | Binary outcome |
|---|---|---|
| Two independent groups | `t.test(y~g)` (Welch); Wilcoxon rank-sum | `prop.test` / `fisher.test` |
| Paired / pre-post | `t.test(pre,post,paired =TRUE)`; Wilcoxon signed-rank | McNemar |
| $>2$ groups, continuous | `aov(y~g)`; Kruskal-Wallis | chi-square |

## 27.30 Effect sizes

| Statistic | What it reports |
|---|---|
| Mean difference + 95% CI | primary for continuous |
| Cohen's $d$ | standardised mean difference |
| Hedges' $g$ | small-sample correction of $d$ |
| Risk ratio (RR) / odds ratio (OR) | two proportions |
| Risk difference | absolute, clinically intuitive |

```
effectsize::cohens_d(y ~ g)
```

## 27.31 Two proportions

```r
prop.test(c(tA, tB), c(nA, nB))         # asymptotic
fisher.test(matrix(c(tA, nA - tA,
                     tB, nB - tB), 2))  # exact, small cells
```

Report RR (or OR) with 95% CI, not just the $p$-value.

## 27.32 Correlation

| Method | Captures | Assumptions |
|---|---|---|
| Pearson | linear association, continuous | bivariate normal, no outliers |
| Spearman | monotonic association | rank-based, robust |
| Kendall | concordant/discordant pairs | robust, slow on large data |

```r
cor.test(x, y, method = "spearman")
```

## 27.33 Non-parametric tests

| Test | Replaces |
|---|---|
| Wilcoxon rank-sum (Mann-Whitney) | two-sample $t$ |
| Wilcoxon signed-rank | paired $t$ |
| Kruskal-Wallis | one-way ANOVA |
| Sign test | paired $t$, when even ranks fail |

## 27.34 Power and sample size

```r
pwr::pwr.t.test(d = 0.5, power = 0.80, sig.level = 0.05,
                type = "two.sample")
```

| Family | Function |
|---|---|
| two-sample $t$ | `pwr.t.test` |
| two proportions | `pwr.2p.test`, `pwr.2p2n.test` |
| correlation | `pwr.r.test` |
| one-way ANOVA | `pwr.anova.test` |

Simulation-based power for anything the textbooks skip: `simr::powerSim`.

## 27.35 Reporting with `gtsummary`

```
library(gtsummary)
trial |>
  tbl_summary(by = arm, statistic = list(all_continuous() ~ "{mean}
   ↪  ({sd})")) |>
  add_p() |>
  add_overall()
```

## 27.36 Decision rule

- Ask: one-group, two-group, paired, or many-group?
- Check: normal-ish, or should I use a rank-based test?
- Report: point estimate + 95% CI + effect size; $p$-value last, not first.

## 27.37 Common pitfalls

- Equal-variance $t$-test (default in some languages) when variances differ.
- Reporting OR when the audience expects RR (or vice versa).
- Forgetting that the $p$-value of a paired test depends on the pairing.
- Chaining correlations until one is "significant".

## 27.38 Further reading

- Altman, Machin et al., *Statistics with Confidence*.
- `gtsummary` docs.

## 27.39 Regression

### 27.39.1 Linear regression and diagnostics

#### 27.39.1.1 Correlation vs regression

- Correlation: symmetric; both variables random (**Model II**).
- Regression: asymmetric; predictor fixed, outcome random (**Model I**).
- Use Model II (e.g. `smatr::sma`) when both variables have error.

## 27.40 Simple linear regression

$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$

```
fit <- lm(y ~ x, data = df)
broom::tidy(fit, conf.int = TRUE)
```

```
broom::glance(fit)
```

## 27.41 Multiple regression

```
fit <- lm(y ~ x1 + x2 + x1:x2, data = df)
```

| Concept | Fix |
|---|---|
| Confounding | adjust by including the confounder |
| Interaction | include `x1:x2` (or `x1*x2`) |
| Non-linearity | `splines::ns(x,4)` or `I(x^2)` |
| Scale sensitivity | centre / standardise predictors |

## 27.42 Diagnostics

```
plot(fit)                    # base: 4 diagnostic plots
performance::check_model(fit)   # tidy overview
car::vif(fit)                # multicollinearity
```

| Plot | What it flags |
|---|---|
| Residuals vs fitted | non-linearity, non-constant variance |
| Normal Q-Q | non-normal residuals |
| Scale-location | heteroscedasticity |
| Leverage / Cook's | influential outliers |

VIF $> 5 \rightarrow$ investigate collinearity; $> 10 \rightarrow$ fix it.

## 27.43 Robust / weighted

```
MASS::rlm(y ~ x, data = df)          # robust to outliers
lm(y ~ x, weights = 1 / var_i)        # weighted LS

# Heteroscedasticity-consistent SEs
library(sandwich); library(lmtest)
coeftest(fit, vcov = vcovHC(fit, type = "HC3"))
```

## 27.44 Decision rule

- Residual plot ugly? Transform, add terms, or switch link.

- VIF explodes? Drop one of the collinear predictors or combine them.
- Outlier with high Cook's distance? Investigate before deleting.
- Variance depends on X? HC3 SEs or weighted LS.

## 27.45 Common pitfalls

- Interpreting $\beta_1$ as "effect of X" under confounding.
- Reporting $R^2$ on in-sample data as if it were generalisable.
- Including an interaction but only reporting main effects.
- Centring categorical predictors (don't — use reference coding).

## 27.46 Further reading

- Harrell, *Regression Modeling Strategies*, ch. 4–5.
- Fox, *Applied Regression Analysis*.

### 27.46.1 ANOVA, mixed models, and non-linear regression

#### 27.46.1.1 One-way ANOVA

```
aov(y ~ group, data = df) |> summary()
```

ANOVA is a linear model with a categorical predictor. The F-test compares between-group to within-group variance.

## 27.47 Contrasts with `emmeans`

```
library(emmeans)
emm <- emmeans(fit, ~ group)
pairs(emm, adjust = "tukey")   # all pairwise
contrast(emm, list(TrtVsCtrl = c(-1, 1, 1, 1) / 3))
```

Pre-specify contrasts before looking at the data; correct for multiplicity.

## 27.48 Two-way / factorial ANOVA

```
aov(y ~ A * B, data = df) |> summary()
emmip(fit, A ~ B)   # interaction plot
```

Interaction means "effect of A differs by level of B". Report the interaction first; main effects are conditional.

## 27.49 Repeated measures / blocking

- RCBD: `aov(y~treatment+Error(block))`.
- Repeated measures: move to a mixed model.

```
library(lme4); library(lmerTest)
lmer(y ~ treatment + time + (1 | subject), data = df)
```

## 27.50 GAMs — smooth non-linear terms

```
library(mgcv); library(gratia)
fit <- gam(y ~ s(x, k = 10) + z, data = df)
summary(fit)      # edf tells you how "wiggly"
draw(fit)         # smooth + CI
```

edf $\approx 1 \rightarrow$ nearly linear; $> 4 \rightarrow$ clearly non-linear.

## 27.51 Non-linear regression (`nls`)

```
# Michaelis-Menten: y = Vmax * x / (K + x)
fit <- nls(y ~ Vmax * x / (K + x),
           data = df, start = list(Vmax = 1, K = 1))
```

Start values matter. If it fails, plot first to guess reasonable starts.

## 27.52 Decision rule

- Categorical predictor, $> 2$ levels $\rightarrow$ ANOVA + contrasts.
- Factorial design $\rightarrow$ include interaction, report it first.
- Effect obviously curved $\rightarrow$ GAM with spline; else try `nls`.
- Repeated measures $\rightarrow$ mixed model, not repeated-measures ANOVA.

## 27.53 Common pitfalls

- Tukey HSD without pre-specified contrasts of interest.
- ANOVA $p < 0.05$ reported alone — without naming *which* groups differ.
- Forcing a GAM onto monotonic data that `nls` fits cleanly.
- Ignoring the random effect in clustered designs (pseudoreplication).

## 27.54 Further reading

- Wood, *Generalized Additive Models*, 2e.

- `emmeans` vignette.

### 27.54.1 Generalised linear models and prediction evaluation

#### 27.54.1.1 GLM link functions

| Outcome | Family | Link | R |
|---|---|---|---|
| Continuous | gaussian | identity | `lm()` |
| Binary | binomial | logit | `glm(y~x,family=binomial)` |
| Count | poisson | log | `glm(y~x,family=poisson,offset=log(t))` |
| Overdispersed count | neg. binomial | log | `MASS::glm.nb` |
| Ordinal | cumulative | logit | `MASS::polr` |
| Nominal | multinomial | logit | `nnet::multinom` |

## 27.55 Logistic regression

```r
fit <- glm(y ~ x1 + x2, data = df, family = binomial)
exp(coef(fit))                       # odds ratios
exp(confint(fit))                    # 95% CI on OR
predict(fit, newdata = nd, type = "response")
```

- Check for perfect separation (huge SEs).
- Interpret ORs cautiously; RR is more intuitive for the audience.

## 27.56 ANCOVA in an RCT

Adjust for baseline; **do not** analyse the change score.

```r
lm(y_followup ~ arm + y_baseline, data = trial)
```

More efficient than simple t-test on change scores when baseline and follow-up are correlated.

## 27.57 Poisson / negative binomial

```r
glm(cases ~ x + offset(log(person_years)),
    family = poisson, data = df)
MASS::glm.nb(cases ~ x + offset(log(person_years)), data = df)
```

Check dispersion: `sum(residuals(fit,type="pearson")^2)/df.residual`. If > 1.5, switch to NB.

## 27.58 Evaluation — calibration + discrimination

| Metric | Means | R |
|---|---|---|
| Calibration plot | predicted vs observed | `rms::val.prob`, manual bin |
| ROC / AUC | rank ordering | `pROC::roc(y,phat)` |
| Brier score | overall accuracy | `mean((phat-y)^2)` |
| Calibration slope / intercept | systematic bias | from logistic recalibration |

```r
library(pROC)
roc_obj <- roc(y, phat)
auc(roc_obj); ci.auc(roc_obj)
plot(roc_obj)
```

## 27.59 Decision rule

- Binary outcome → logistic; report OR + 95% CI.
- Count outcome → Poisson; check overdispersion; NB if needed.
- Trial analysis → ANCOVA, not change score.
- Prediction model → calibration curve first, ROC second, decision curves third.

## 27.60 Common pitfalls

- Quoting AUC without calibration (a discriminating but miscalibrated model is dangerous).
- Ignoring offsets in count data.
- Using ordinal logit when the proportional-odds assumption fails.
- Presenting logistic regression coefficients on the log-odds scale without OR.

## 27.61 Further reading

- Harrell, *RMS*, ch. 10–12.
- Steyerberg, *Clinical Prediction Models*.

### 27.61.1 Agreement, survival primer, and reporting

#### 27.61.1.1 Don't dichotomise

Cutting a continuous variable at the median loses $\approx 37\%$ of the information for a linear association. Keep predictors continuous; model non-linearity with a spline if needed.

## 27.62 Change scores vs ANCOVA

- Change scores suffer from **regression to the mean**.
- ANCOVA (adjust for baseline) is the efficient analysis in an RCT.
- In observational studies with unequal baselines, both approaches answer subtly different questions — state which.

## 27.63 Agreement & reliability

| Statistic | Use | R |
|---|---|---|
| Cohen's $\kappa$ | two raters, categorical | `irr::kappa2(cbind(r1,r2))` |
| Weighted $\kappa$ | ordinal categorical | `irr::kappa2(...,weight="squared")` |
| ICC(2,1) / ICC(3,1) | continuous | `psych::ICC(df)` |
| Bland-Altman | two methods, continuous | plot $\bar{xy}$ vs $y - x$ |

```r
mean_diff <- mean(y - x)
loa <- mean_diff + c(-1.96, 1.96) * sd(y - x)
ggplot(df, aes((x + y)/2, y - x)) + geom_point() +
  geom_hline(yintercept = c(mean_diff, loa), linetype = 2)
```

## 27.64 Survival primer

| Concept | R |
|---|---|
| Kaplan-Meier | `survfit(Surv(time,event)~g)` + `ggsurvfit::ggsurvfit` |
| Log-rank test | `survdiff(Surv(time,event)~g)` |
| Cox PH model | `coxph(Surv(time,event)~x1+x2)` |
| PH check | `cox.zph(fit)` |

- Report a hazard ratio with 95% CI.

- PH violated? → time-varying coefficient or stratify.
- Interpret HR only after checking for non-informative censoring.

## 27.65 Decision curves, NRI, IDI

- **Decision curve**: net benefit vs threshold probability — dominates "treat all" and "treat none" when useful.
- **NRI**: how many events move up / non-events move down in risk.
- **IDI**: mean change in predicted probability by class.
- Report decision curve first; NRI/IDI as secondary.

## 27.66 Explanation vs prediction (Shmueli)

| Explanatory | Predictive |
|---|---|
| Goal: inference about $\beta$ | Goal: minimise out-of-sample loss |
| Tools: ANOVA, diagnostics, intervals | Tools: CV, regularisation, ensembles |
| Metric: interval coverage, bias | Metric: RMSE, AUC, Brier on hold-out |

## 27.67 Reporting guidelines

| Design | Guideline |
|---|---|
| Randomised trial | CONSORT |
| Observational | STROBE |
| Diagnostic-accuracy | STARD |
| Prediction model (incl. AI) | TRIPOD / TRIPOD-AI |
| Systematic review | PRISMA |
| Animal research | ARRIVE |

## 27.68 Decision rule

- Continuous predictor + continuous outcome → no median splits.
- Two-group with repeated measurement → ANCOVA.
- New rater / device → Bland-Altman + ICC.
- Time-to-event → KM + Cox, always check PH.
- Prediction model → TRIPOD checklist at submission.

## 27.69 Common pitfalls

- Reporting $\kappa$ on near-constant outcomes (high % agreement, low $\kappa$).

- Quoting median survival that is never reached.
- Forgetting to mention PH assumption checking.
- Claiming a "prediction model" built on the same data used for evaluation.

## 27.70 Further reading

- Harrell, *BBR*, ch. 17–18.
- Royston & Altman, *Prognosis research*.

## 27.71 Design & Causal Inference

### 27.71.1 Study designs and power

#### 27.71.1.1 Observational designs

| Design | Strengths | Weaknesses |
|---|---|---|
| Cohort | temporality, incidence | expensive, loss to follow-up |
| Case-control | rare outcomes, cheap | recall + selection bias |
| Cross-sectional | prevalence, screening | no temporality |
| Case-crossover | within-person comparison | for transient exposures only |

STROBE checklist: https://www.strobe-statement.org/

## 27.72 Trial designs

| Type | When |
|---|---|
| Parallel-group | most common; independent arms |
| Crossover | stable chronic condition, carry-over washable |
| Cluster | intervention at group level (schools, clinics) |
| Factorial | two or more interventions, test interactions |
| Adaptive | pre-specified modifications based on interim data |
| Non-inferiority | new treatment not worse than control by margin $\Delta$ |
| Equivalence | two-sided non-inferiority |

CONSORT for RCTs: https://www.consort-statement.org/

## 27.73 Bench / translational

- **Blocking** reduces nuisance variation (plate, day, operator).
- **Factorial** tests interactions efficiently.
- **Split-plot** handles two levels of randomisation.
- **Pseudoreplication**: technical replicates ≠ biological replicates.

## 27.74 Power — closed form

```r
library(pwr)
pwr.t.test(d = 0.5, power = 0.80, sig.level = 0.05,
           type = "two.sample")
pwr.2p.test(h = ES.h(0.3, 0.2), power = 0.80)
pwr.r.test(r = 0.3, power = 0.80)
pwr.anova.test(k = 4, f = 0.25, power = 0.80)
```

Effect-size conventions (Cohen): small $d = 0.2$, medium 0.5, large 0.8.

## 27.75 Power — simulation

```r
library(simr)
# Build a pilot model, increase N, or tweak fixef, then:
ps <- powerSim(model, nsim = 500, test = fixed("arm"))
ps
```

Simulation wins for any design the textbook skips: mixed models, adaptive rules, non-standard outcomes.

## 27.76 Decision rule

- Randomise if you can. If not, draw a DAG and name the biases.
- Power calculation before the protocol freeze, not after data collection.
- Cluster randomisation → design effect $1 + (\bar{m} - 1)\rho$; inflate N.
- Bench experiments → treat batch, plate, and operator as random effects.

## 27.77 Common pitfalls

- Planning a cluster RCT without inflating N for design effect.
- Borrowing a pilot effect size without acknowledging noise.
- Running a non-inferiority trial as if it were superiority.
- Treating technical replicates as if they were biological.

## 27.78 Further reading

- Matthews, *Introduction to Randomized Controlled Clinical Trials*.
- Chow & Liu, *Design and Analysis of Clinical Trials*.

### 27.78.1 Missing data, mixed models, and longitudinal

#### 27.78.1.1 Missingness mechanisms

| Mechanism | Definition | Complete-case analysis? |
|---|---|---|
| MCAR | missingness independent of data | unbiased but inefficient |
| MAR | missingness depends on **observed** data | biased, imputation helps |
| MNAR | missingness depends on **unobserved** values | imputation under model only |

Test is impossible without assumptions — reason about the mechanism.

## 27.79 Multiple imputation with `mice`

```r
library(mice)
imp <- mice(df, m = 20, seed = 42, printFlag = FALSE)
fit <- with(imp, lm(y ~ x1 + x2))
pooled <- pool(fit)
summary(pooled, conf.int = TRUE)
```

Rule of thumb: $m \geq 100 \times$ fraction of missing information, but 20 is a fine starting point.

## 27.80 Linear mixed models

```r
library(lme4); library(lmerTest)
lmer(y ~ treatment * time + (1 | subject), data = df)
lmer(y ~ treatment * time + (time | subject), data = df)  # random slope
```

- Random intercept: each cluster has its own baseline.
- Random slope: each cluster has its own trajectory.
- ICC $\rho = \sigma_u^2 / (\sigma_u^2 + \sigma_e^2)$.

REML for variance components; ML only for likelihood-ratio tests on fixed effects.

## 27.81 GLMMs & GEE

```r
library(lme4)
glmer(y ~ x + (1 | cluster), data = df, family = binomial)

library(geepack)
geeglm(y ~ x, id = cluster, data = df,
       family = binomial, corstr = "exchangeable")
```

- GLMM: subject-specific effects, good for prediction.
- GEE: population-average effects, robust to working-correlation misspecification (when combined with robust SEs).

## 27.82 Time series basics

```r
library(forecast)
ts_obj <- ts(x, frequency = 12)
decompose(ts_obj) |> plot()
auto.arima(ts_obj)

library(changepoint)
cpt.mean(x, method = "PELT") |> plot()
```

## 27.83 Decision rule

- Missing data $> 5\%$ $\rightarrow$ at least sensitivity analysis; prefer MI.
- Repeated measurements $\rightarrow$ mixed model, not repeated-measures ANOVA.
- Binary outcome with clusters $\rightarrow$ GLMM (if subject effects matter) or GEE (if marginal effect is the estimand).
- Time series with trend / seasonality $\rightarrow$ decompose before modelling.

## 27.84 Common pitfalls

- Last-observation-carried-forward; usually biased.
- Imputing after transformation rather than in the right scale.
- Ignoring random effects when clusters are small in number (degrees of freedom correction).
- Quoting GEE coefficients as if they were subject-specific.

## 27.85 Further reading

- van Buuren, *Flexible Imputation of Missing Data*.

- Fitzmaurice, Laird, Ware, *Applied Longitudinal Analysis*.

### 27.85.1 Causal inference methods

#### 27.85.1.1 Time-varying covariates / landmarking

- Counting-process data: split follow-up into intervals.
- Landmark analysis: define a landmark time $t^*$, condition on surviving to $t^*$, classify by status at $t^*$.

```
library(survival)
coxph(Surv(tstart, tstop, event) ~ x, data = long)
```

- Immortal-time bias: exposed group can only be exposed if they live long enough to be exposed → spurious survival benefit.

## 27.86 Competing risks

```
library(tidycmprsk); library(ggsurvfit)
cif <- cuminc(Surv(time, event_f) ~ x, data = df)
ggsurvfit::ggcuminc(cif, outcome = "event1")

# Fine-Gray subdistribution hazard for event1
crr(Surv(time, event_f) ~ x, data = df, failcode = "event1")
```

- Cause-specific HR answers "given you are event-free, what is the hazard for event $k$?"
- Subdistribution HR answers "what is the hazard for event $k$ in the whole cohort, treating competing events as obstacles?"

## 27.87 DAGs

```
library(dagitty); library(ggdag)
dag <- dagitty("dag { X -> Y ; C -> X ; C -> Y }")
adjustmentSets(dag, exposure = "X", outcome = "Y")
impliedConditionalIndependencies(dag)
```

- Adjust for confounders, **not** colliders.
- Adjusting for a mediator estimates the direct effect only.

## 27.88 Propensity scores / IPTW

```
library(MatchIt); library(cobalt)
m <- matchit(treat ~ x1 + x2, data = df, method = "nearest")
```

```r
love.plot(m)   # check balance: SMD < 0.1 is the goal
fit <- lm(y ~ treat, data = match.data(m))
```

IPTW: weight by $1/\hat{e}_i$ for treated, $1/(1-\hat{e}_i)$ for untreated. Stabilised weights stay closer to 1; trim extremes to avoid outliers.

## 27.89 G-methods, IV, DiD, RDD

| Method | Identifying assumption |
|---|---|
| G-formula / IPTW | positivity + no unmeasured confounding |
| Instrumental variables | exclusion restriction + relevance |
| Difference-in-differences | parallel trends |
| Regression discontinuity | continuity at the cutoff |

## 27.90 Heterogeneous treatment effects

- Causal forest (`grf::causal_forest`) or meta-learners (S-, T-, X-).
- Report conditional ATEs with adjusted intervals; avoid spurious subgroup p-values.

## 27.91 Decision rule

- Treatment assigned over time → time-varying Cox.
- Multiple competing events → cumulative incidence, Fine-Gray.
- Observational causal question → DAG first, choose identification strategy second.
- Want the whole distribution of treatment effect? → causal forest.

## 27.92 Common pitfalls

- Adjusting for a post-treatment variable (induces collider bias).
- Quoting HR from a Cox model that violates PH.
- Matching but never checking balance.
- Calling IV estimates "generalisable" when they are local to compliers.

## 27.93 Further reading

- Hernán & Robins, *Causal Inference: What If*.
- Therneau & Grambsch, *Modeling Survival Data*.

### 27.93.1 Synthesis and pre-registration

#### 27.93.1.1 Systematic review workflow

1. PICO: Population, Intervention, Comparator, Outcome.
2. Register protocol on PROSPERO.
3. Build search across $\geq 2$ databases; peer-review with a librarian.
4. Two-screener title/abstract and full-text.
5. Extract, appraise (RoB 2 / ROBINS-I), synthesise.
6. Report with the PRISMA 2020 checklist and flow diagram.

## 27.94 Meta-analysis

```r
library(metafor)
dat <- escalc(measure = "RR",
              ai = tpos, bi = tneg, ci = cpos, di = cneg,
              data = studies)
fit <- rma(yi, vi, data = dat, method = "REML")
fit
forest(fit, header = TRUE)
funnel(fit); regtest(fit)
```

| Statistic | Meaning |
|---|---|
| $\tau^2$ | between-study variance |
| $I^2$ | % of total variance due to heterogeneity |
| Egger's test | funnel-plot asymmetry (small-study effects) |

Fixed-effect only when heterogeneity is effectively zero. Default to random effects.

## 27.95 Network meta-analysis

```r
library(netmeta)
net <- netmeta(TE, seTE, treat1, treat2, studlab,
               data = dat, sm = "MD", random = TRUE)
netgraph(net)
netrank(net, small.values = "good")    # SUCRA ranking
decomp.design(net)                     # inconsistency
```

Transitivity + consistency are the core assumptions.

## 27.96 Infectious disease — SIR / SEIR

```r
library(deSolve)
sir <- function(t, y, p) with(as.list(c(y, p)), {
  N <- S + I + R
  list(c(-beta * S * I / N,
          beta * S * I / N - gamma * I,
          gamma * I))
})
out <- ode(c(S = 999, I = 1, R = 0),
           times = 0:120,
           func = sir,
           parms = c(beta = 0.3, gamma = 0.1))
```

$R_0 = \beta/\gamma$. Herd-immunity threshold $\approx 1 - 1/R_0$.

## 27.97 Pre-registration and SAP

A pre-registration specifies, before data analysis:

- Primary outcome + exact estimand.
- Analysis model (fixed terms, adjustments, interactions).
- Handling of missing data.
- Multiplicity control.
- Sample-size justification.

OSF and ClinicalTrials.gov are the two common registries.

## 27.98 Decision rule

- More than a handful of studies on a question → plan a systematic review.
- Multiple comparators of interest → network meta-analysis.
- Outbreak modelling → start with SIR, extend only as data justify.
- Confirmatory analysis → pre-registered; exploratory analyses are labelled as such in the manuscript.

## 27.99 Common pitfalls

- Declaring a review “systematic” without a pre-registered protocol.
- Pooling studies with fundamentally different populations / outcomes.
- Interpreting SUCRA rankings as if they were certain.
- Calling a compartmental model a “forecast” without uncertainty.

## 27.100 Further reading

- Higgins et al., *Cochrane Handbook for Systematic Reviews*.
- Keeling & Rohani, *Modeling Infectious Diseases in Humans and Animals*.
- Nosek et al. (2018), *The preregistration revolution*.

## 27.101 Machine Learning & High-Dimensional Data

### 27.101.1 Cross-validation, regularisation, and unsupervised

#### 27.101.1.1 Cross-validation

| Scheme | Use |
|---|---|
| k-fold (k = 5 or 10) | baseline, fast |
| Repeated k-fold | reduces Monte Carlo noise |
| Leave-one-out | small n, high variance |
| Stratified k-fold | imbalanced class labels |
| Grouped / blocked | correlated units (patients, sites) |
| Nested | honest evaluation of tuned models |

Nested CV is the only defensible way to estimate generalisation error when you are also tuning hyperparameters.

## 27.102 Regularisation

```r
library(glmnet)
x <- model.matrix(y ~ . - 1, df); y <- df$y
fit <- cv.glmnet(x, y, alpha = 1)     # alpha = 1 = lasso; 0 = ridge
coef(fit, s = "lambda.1se")           # sparse solution
plot(fit)                              # CV curve
```

- `lambda.min`: empirical minimum of CV error.
- `lambda.1se`: more regularised, simpler, within 1 SE of min.
- Elastic net: `alpha` between 0 and 1 blends ridge and lasso.

## 27.103 Multivariate workhorses

| Method | Purpose |
|---|---|
| PCA | variance → low-d summary; `prcomp(x,scale.=TRUE)` |
| FA | latent-variable modelling; `psych::fa` |
| CCA | correlation structure between two sets |
| LDA | supervised low-d projection; `MASS::lda` |

```
pc <- prcomp(x, scale. = TRUE)
summary(pc)   # proportion of variance per PC
```

Scale before PCA when variables are on different units.

## 27.104 Clustering

```
km <- kmeans(scale(x), centers = 4, nstart = 25)
hc <- hclust(dist(scale(x)), method = "ward.D2")
library(mclust); bic <- mclustBIC(x)      # model-based, BIC-selected
```

- Choose k by silhouette, gap statistic, or BIC.
- Scale first. K-means assumes round clusters of similar size.

## 27.105 Non-linear embeddings

```
library(uwot)
umap_xy <- umap(scale(x), n_neighbors = 15, min_dist = 0.1)

library(Rtsne)
ts <- Rtsne(scale(x), perplexity = 30)$Y
```

Embeddings are *visualisations*, not distances you should input into downstream stats. Global geometry is often distorted.

## 27.106 Decision rule

- Lots of predictors, small n → lasso; report `lambda.1se`.
- Exploring structure → PCA + UMAP on scaled data.
- Clustering for biology → model-based + sensitivity to k.
- Tuning anything → nested CV.

## 27.107 Common pitfalls

- Selecting features on full data, then estimating error by k-fold.

- PCA before scaling when features are on different units.
- Interpreting UMAP distances as real distances.
- Using AUC on rank-ordered hold-out error as if it were CV error.

## 27.108 Further reading

- Hastie, Tibshirani, Friedman, *The Elements of Statistical Learning*.
- James et al., *ISLR2*.

### 27.108.1 Trees, boosting, and interpretability

#### 27.108.1.1 Tree-based models

| Model | R | Strengths |
|---|---|---|
| CART | `rpart::rpart` | interpretable single tree |
| Random forest | `ranger::ranger` | robust, OOB error, variable importance |
| XGBoost | `xgboost::xgb.train` | top tabular performance, careful tuning |
| LightGBM | `lightgbm::lgb.train` | fast on large data |

```r
library(ranger)
fit <- ranger(y ~ ., data = df, importance = "permutation",
              num.trees = 500, mtry = floor(sqrt(ncol(df) - 1)))
fit$prediction.error   # OOB error
```

## 27.109 Interpretability

```r
library(DALEX); library(iml)

ex  <- explain(fit, data = df |> select(-y), y = df$y)
ip  <- model_parts(ex)           # permutation importance
pd  <- model_profile(ex, variables = "x1")   # partial dependence
sh  <- Shapley$new(Predictor$new(fit, data = df, y = df$y),
                   x.interest = df[1, ])     # SHAP
```

PDP assumes feature independence; if features are correlated, use ALE plots.

## 27.110 Tabular neural networks with `torch`

```
# sketch only — typically in a loop on GPU
library(torch)
nn <- nn_sequential(
  nn_linear(p, 64), nn_relu(),
  nn_linear(64, 1)
)
opt <- optim_adam(nn$parameters)
```

On small biomedical tabular data, boosted trees usually beat NN. Use NN when you need end-to-end training with images, sequences, or text.

## 27.111 tidymodels pipeline

```
library(tidymodels)
rec  <- recipe(y ~ ., data = df) |>
  step_normalize(all_numeric_predictors()) |>
  step_dummy(all_nominal_predictors())

mod  <- rand_forest(mtry = tune(), trees = 500) |>
  set_engine("ranger") |> set_mode("classification")

wf   <- workflow() |> add_recipe(rec) |> add_model(mod)
cv   <- vfold_cv(df, v = 5, strata = y)
tuned <- tune_grid(wf, cv, grid = 20)
collect_metrics(tuned)
```

## 27.112 Decision rule

- Tabular, $< 100$k rows $\rightarrow$ gradient-boosted trees.
- Need explanations $\rightarrow$ SHAP + PDP, validate with held-out.
- Images / sequences $\rightarrow$ CNN / transformer, use `torch`.
- Reproducibility $\rightarrow$ `tidymodels` pipeline + fixed seed + locked recipe.

## 27.113 Common pitfalls

- Reporting feature importance from a model trained on the full data.
- Interpreting PDPs when features are strongly correlated.
- Claiming NN superiority without CV on the same split as a tree model.
- Forgetting to set a seed for any tune / split operation.

## 27.114 Further reading

- Biecek & Burzykowski, *Explanatory Model Analysis*.
- Molnar, *Interpretable Machine Learning*.

### 27.114.1 Bayesian and survival ML

#### 27.114.1.1 Bayesian thinking

$$\underbrace{p(\theta \mid y)}_{\text{posterior}} \propto \underbrace{p(y \mid \theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}$$

- Priors are part of the model — specify and defend them.
- Posterior summaries: mean, median, 95% credible interval.
- No $p$-values; use posterior probability of direction, Bayes factors, or LOO.

## 27.115 `brms` / Stan

```
library(brms)
fit <- brm(y ~ x + (1 | group), data = df,
           prior = c(prior(normal(0, 1), class = "b"),
                     prior(exponential(1), class = "sd")),
           chains = 4, iter = 2000, seed = 42)

summary(fit)
loo(fit)               # leave-one-out cross-validation
pp_check(fit)           # posterior predictive checks
```

Prior predictive check before fitting: simulate $y$ from the prior and confirm the implications are plausible.

## 27.116 Biomarker statistics

| Question | Statistic |
|---|---|
| Does a biomarker classify? | AUC, cut-point (Youden's index) |
| Does it add over an existing model? | ΔAUC, NRI, IDI, decision curves |
| Does it move prognosis? | Calibration plus discrimination |
| Is a cut-off reproducible? | Bootstrap CI on the cut-off |

```
pROC::roc(y, biomarker) |>
  pROC::coords("best", best.method = "youden")
```

## 27.117 Survival ML

| Model | R |
|---|---|
| Random survival forest | `randomForestSRC::rfsrc(Surv(...)~., data)` |
| Gradient-boosted Cox | `xgboost` with `objective="survival:cox"` |
| DeepSurv (conceptual) | `torch` with partial-likelihood loss |

Evaluate with **time-dependent AUC**, **integrated Brier score**, and **IPA** (index of prediction accuracy = 1 − Brier(model) / Brier(null)).

```
library(timeROC)
r <- timeROC(T = df$time, delta = df$event,
             marker = pred, cause = 1, times = c(365, 730))
```

## 27.118 External validation

- Never trust the apparent performance on training data.
- Minimum: split-sample or CV on internal data.
- Better: external validation in a second cohort.
- Report calibration slope / intercept, discrimination, and decision curves.

## 27.119 Decision rule

- Rare outcome with need for uncertainty → Bayesian, not bootstrap of MLE.
- New biomarker → NRI + decision curve, not just ΔAUC.
- Survival prediction → time-dependent Brier / IPA, not global AUC.
- Any clinical claim → external validation before publication.

## 27.120 Common pitfalls

- Reporting a posterior with default flat priors on unscaled predictors.
- Selecting the Youden cut-off on the full data and using the same data to evaluate sensitivity / specificity.
- Quoting C-statistic at a single time point and calling it survival ML.
- Publishing a prediction model without TRIPOD-compliant reporting.

## 27.121 Further reading

- Gelman et al., *Bayesian Data Analysis*, 3e.

- Royston & Altman, *External validation of a Cox prognostic model.*

### 27.121.1 Omics, FDR, and TRIPOD-AI

#### 27.121.1.1 Bulk RNA-seq

```
library(DESeq2)
dds <- DESeqDataSetFromMatrix(counts, coldata, ~ condition)
dds <- DESeq(dds)
res <- results(dds, contrast = c("condition", "B", "A"))
```

```
library(edgeR)
dge <- DGEList(counts, group = coldata$condition)
dge <- calcNormFactors(dge)
design <- model.matrix(~ condition, coldata)
fit <- glmQLFit(estimateDisp(dge, design), design)
qlf <- glmQLFTest(fit, coef = 2)
topTags(qlf)
```

| Package | Distribution | Notes |
|---|---|---|
| DESeq2 | negative binomial | shrinks LFCs; standard for small cohorts |
| edgeR | negative binomial | QL-F test has good type-I control |
| limma-voom | mean-variance weights | fast on very large studies |

## 27.122 Enrichment

```
library(fgsea)
fgsea(pathways, stats = stat_vector, nperm = 10000) |>
  arrange(padj)

# Over-representation with a hypergeometric test
phyper(k - 1, K, N - K, n, lower.tail = FALSE)
```

Always pre-rank by a *signed* statistic (log-FC × -log10 p), not by raw p alone.

## 27.123 Single-cell RNA-seq (Seurat pattern)

```
library(Seurat)
so <- CreateSeuratObject(counts) |>
```

```
  NormalizeData() |>
  FindVariableFeatures() |>
  ScaleData() |>
  RunPCA() |>
  FindNeighbors() |>
  FindClusters(resolution = 0.4) |>
  RunUMAP(dims = 1:20)
DimPlot(so)
```

## 27.124 FDR, knockoffs, replication

| Correction | When |
|---|---|
| Bonferroni | few planned tests, strict control |
| Benjamini-Hochberg | many tests, accept proportion of false discoveries |
| Knockoffs | controlled variable selection with FDR |
| Permutation / SEA | small $n$, non-standard models |

```
p.adjust(pvals, method = "BH")
```

Replication crisis shorthand: pre-register + report effect sizes + share data. Power, not p-values.

## 27.125 TRIPOD-AI + fairness

- TRIPOD-AI extends TRIPOD for prediction models built with ML.
- Report: data provenance, training/evaluation splits, subgroup performance, calibration, external validation.
- Fairness: stratify AUC / calibration by sex, ancestry, age bands. Disparities in calibration are often larger than in AUC.

## 27.126 Reproducibility at scale with `targets`

```
# _targets.R
library(targets)
tar_option_set(packages = c("tidyverse"))
list(
  tar_target(raw, read_csv("data/raw.csv")),
  tar_target(clean, clean_data(raw)),
```

```
  tar_target(fit, fit_model(clean)),
  tar_target(report, quarto::quarto_render("report.qmd"))
)
```

```
Rscript -e 'targets::tar_make()'
```

## 27.127 Decision rule

- Bulk DE → DESeq2 for small samples, edgeR/limma for large.
- Many hypotheses → BH by default; knockoffs for variable selection.
- scRNA-seq → Seurat pipeline with batch correction + cluster annotation.
- ML prediction model → TRIPOD-AI + subgroup analysis.
- Any multi-step analysis → `targets` pipeline.

## 27.128 Common pitfalls

- Running DE on a pathway-level aggregate rather than gene-level.
- Cherry-picking a pathway database post-hoc.
- Over-clustering scRNA-seq data until "known" groups appear.
- Calling an ML model "fair" after checking only overall AUC.
- Submitting a paper without a `sessionInfo()` or a lock file.

## 27.129 Further reading

- Love, Huber, Anders (2014), *Moderated estimation of fold change…*
- Collins et al. (2024), *TRIPOD-AI.*
- Peng, *Reproducible Research with R.*